QuantRule — The Distributed Research Platform Kata

QuantRule

2026-11-12

QuantRule — The Distributed Research Platform Kata

About this kata

A one-page, solution-agnostic description of the product, its operator, and its measurable requirements. It is the course submission copy of `docs/architecture/kata.md`; where the two differ, the architecture copy governs.

Name

QuantRule — the reproducible evidence platform.

One line

A distributed research platform that turns an OHLCV hypothesis into reproducible, portfolio-constrained evidence, and keeps that evidence auditable after the client that submitted it has closed.

Intent

Backtests are cheap; trustworthy backtests are not. QuantRule exists so that the same candles and the same configuration produce the same evidence, every time, and so that a negative result is a legitimate outcome. The product is for one operator on an internal network. It is not a public service and it is not a prediction engine.

Operator

A systematic researcher who owns their candles, submits batch runs, and needs the result to survive their laptop closing. A secondary reader is a reviewer who must reproduce a result from an image digest and a canonical digest, without an account.

The product is single-principal. It is not multi-user and does not pretend to be.

Job to be done

Given candle data, a data version, a model, a strategy, and portfolio constraints, submit a run and walk away. Come back to a bundle I can reproduce, inspect, and defend: an audit trail, a canonical digest a reviewer can compare, and a recorded decision to accept the hypothesis, revise it, or stop at "no edge".

Requirements

1. Accept candles as an upload, identify them by content, and keep the bytes.
2. Measure each timeframe with readings, states, confirmed levels, events, progression, and context; publish numbers, nulls, and codes — never labels.
3. Evaluate model, strategy, execution, and portfolio constraints in one deterministic run with explicit costs and no look-ahead.
4. Return `202` on submission and continue after the client disconnects.
5. Produce a bundle with provenance naming the data version, config hash, model version, and image digest, plus a canonical digest over the data artifacts.
6. Expose the same capabilities through a browser client, a command shell, and a desktop shell that reads no local data.
7. Start with one command on a machine with Docker; expose one port.
8. Require no account, and keep candle data on the operator's network.

Constraints and non-goals

- Candles only — no ticks, order books, machine learning, or statistical signals.
- One principal. No accounts, no tenancy, no billing, no cloud provider. Profiles organise research; they do not isolate it.
- No high availability, no failover, no uptime target.
- No public distribution page. Deployment is an image and one command; release artifacts live on GitHub Releases.
- A backtest is not a profitability claim. Out-of-sample evidence is required.

Done when

- One command starts the stack and prints a URL.
- A submitted run survives the client closing and completes.
- A rerun over the same inputs produces the same canonical digest.
- A bundle is refused when its artifact set is incomplete.
- Provenance names the image digest that produced the run.
- Every command, argument, and output shape is discoverable from help.
- A second machine following the verification tutorial prints the same digest.
- A simulation run completes the research loop, premise to decision, in the browser.
- A negative result can be recorded and the workflow can stop.